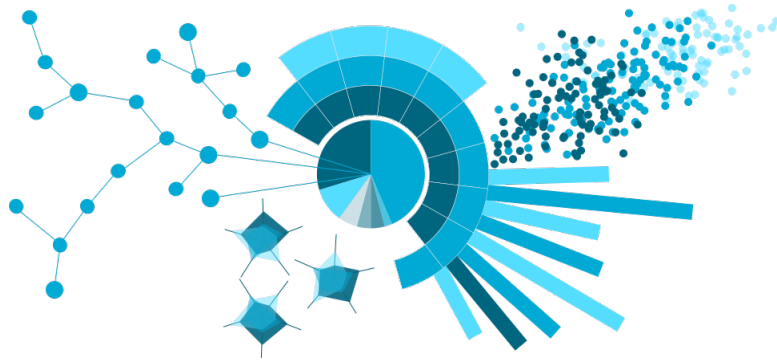


Data Visualization
Assignment 2
Group 4



Ashraf Sarhan
ashsar-5@student.ltu.se
Egil Elenius
egiele-1@student.ltu.se
Zeyad Haroun
zeyhar-5@student.ltu.se
Abdalrahman Dabor
abddab-1@student.ltu.se
Ahmed Selim Al Baker Khan
ahekah-3@student.ltu.se



November 30, 2025

Contents

1	Introduction	3
2	Question 1: US Obesity Map	3
2.1	Question	3
2.2	Data Preprocessing	3
2.3	Visualization	3
2.4	Key Findings	4
3	Question 2: Elevation Map of the USA Using Voronoi Tessellation	4
3.1	Question	4
3.2	Data Preprocessing	4
3.3	Visualization	4
3.4	Key Findings	5
4	Question 3: School Levels Distribution	5
4.1	Question	5
4.2	Data Preprocessing	5
4.3	Visualization	5
4.4	Key Findings	5
5	Question 4: Full-Time Teachers Distribution	6
5.1	Question	6
5.2	Data Preprocessing	6
5.3	Visualization	6
5.4	Key Findings	6
6	Question 5: Enrollment Distribution & Teacher-Student Ratio	7
6.1	Question	7
6.2	Data Preprocessing	7
6.3	Visualization	7
6.4	Key Findings	7
7	Question 6: Global Corruption and Economic Development Analysis	8
7.1	Background	8
7.2	Question	8
7.3	Data Preprocessing	8
7.4	Visualization	8
7.5	Key Findings	9

1 Introduction

The assignment was to answer 6 sets of questions with visualizations and observations for the specified tasks using Jupyter Notebook. Both Preprocessing decisions and insights are summarized for each section.

2 Question 1: US Obesity Map

2.1 Question

Plot a proper map showing the obesity spread over different states of US, highlighting the states with maximum and minimum obesity.

2.2 Data Preprocessing

- Loaded state-level obesity data from `National_Obesity_By_State.csv`.
- Loaded US state boundaries from `US_State.shp` into a `GeoDataFrame`.
- Standardized join keys and merged shapefile `StateName` with CSV column `NAME` (left join).
- Verified `Obesity` as numeric and inspected for missing values (missing entries appear as “No data” on the map).
- Computed the states with maximum and minimum obesity for map highlighting.

2.3 Visualization

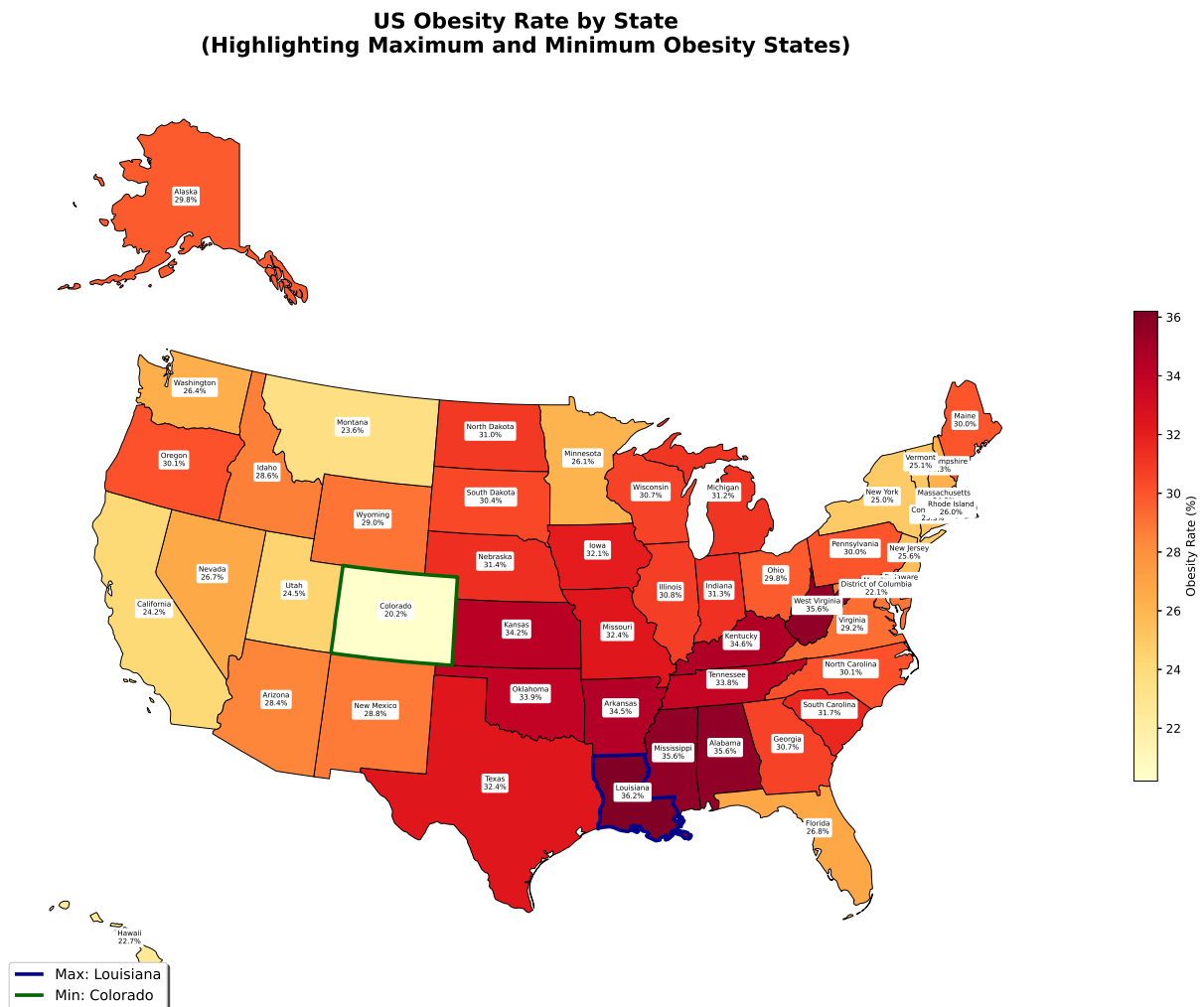


Figure 1: US Obesity Rate by State

This visualization employs a choropleth mapping technique that successfully visualizes obesity rates across US states using shapefile data. The color gradient (yellow to red) effectively shows obesity distribution across geographic areas, demonstrating data smoothing by taking discrete data points (state obesity rates) and spreading them across continuous geographic areas (state boundaries).

2.4 Key Findings

- **Highest Obesity:** Louisiana has the maximum obesity rate at **36.2%** (highlighted with dark blue border)
- **Lowest Obesity:** Colorado has the minimum obesity rate at **20.2%** (highlighted with dark green border)
- **Geographic Pattern:** Southern states tend to have higher obesity rates compared to western and northeastern states
- **Color Scale:** Darker red indicates higher obesity rates, while lighter yellow indicates lower rates
- **Regional Disparity:** 16-percentage point gap between highest and lowest states demonstrates significant regional health inequalities

3 Question 2: Elevation Map of the USA Using Voronoi Tessellation

3.1 Question

Create a Voronoi diagram for the elevations of US cities. Use a data smoothing technique since the elevations are for points, and "spread" those values across areas

3.2 Data Preprocessing

- Loaded elevation point data from ready dataset `usa_cities` (containing latitude, longitude, and elevation values).
- Loaded USA data from ready dataset `contiguous_us` (containing boundaries for each state).
- Converted elevation points into a GeoDataFrame with geographic coordinates.
- Generated Voronoi polygons based on the spatial distribution of the elevation points.
- Clipped the Voronoi polygons to the US boundary shapefile to ensure that polygons do not extend beyond national borders.
- Merged elevation values with the resulting spatial tessellation for visualization.

3.3 Visualization

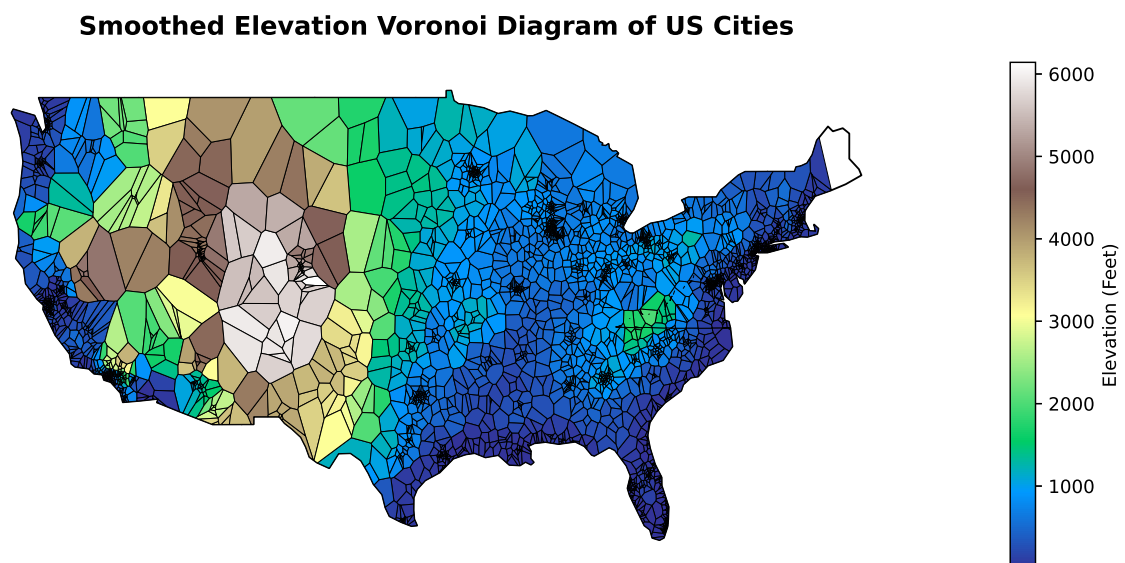


Figure 2: USA Elevation Visualization Using Voronoi Tessellation

This visualization uses a Voronoi tessellation to convert discrete elevation sample points into continuous spatial regions, where each polygon represents the area closest to a given elevation sample. The color gradient reflects elevation intensity, allowing mountain regions and lowland areas to be distinguished spatially.

3.4 Key Findings

- **High Elevation Regions:** The Rocky Mountains and western states exhibit large clusters of high elevation Voronoi cells.
- **Low Elevation Regions:** Eastern and central US areas show predominantly low elevation polygons.
- **Visual Interpretation:** Larger polygons occur where elevation data points are sparse, while smaller polygons indicate dense sampling regions.

4 Question 3: School Levels Distribution

4.1 Question

Plot the distribution of School levels over states and indicate which has the highest and lowest of High and Elementary schools.

4.2 Data Preprocessing

- Loaded public schools point data from `PublicSchools.shp`.
- Grouped by `STATE` and `LEVEL_` to count schools per state-level category.
- Pivoted to wide format for `HIGH` and `ELEMENTARY`; replaced missing counts with 0.
- Mapped state abbreviations to full names via the `us` package to create a `StateName` join key.
- Merged counts with `US_State.shp` polygons for mapping.

4.3 Visualization

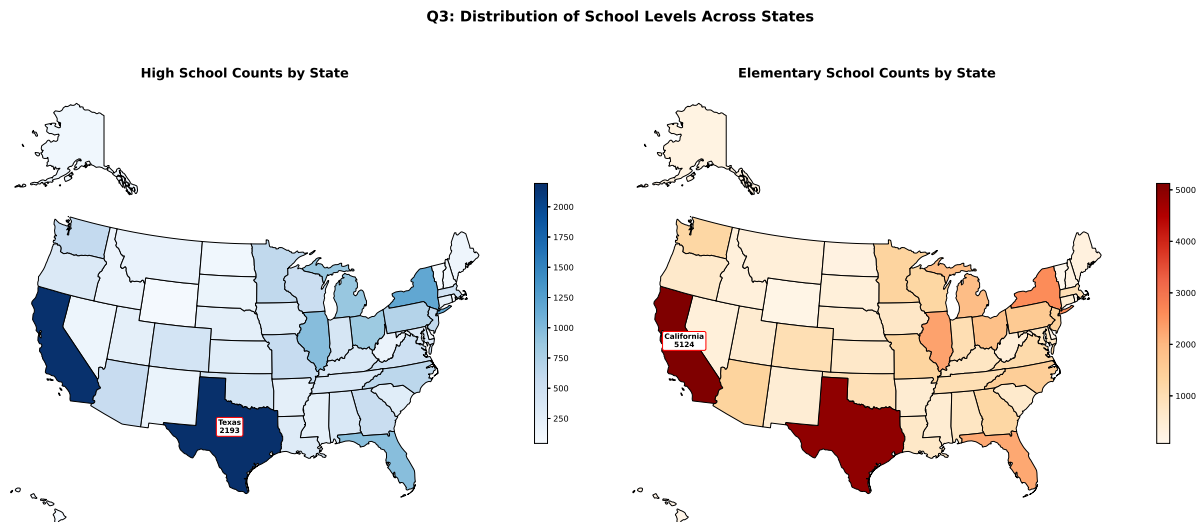


Figure 3: High School Counts (left) and Elementary School Counts (right) by State

4.4 Key Findings

- **High Schools:**
 - Highest: Texas (TX) with 2,193 schools
 - Lowest: Virgin Islands (VI) with 5 schools
- **Elementary Schools:**

- Highest: California (CA) with 5,124 schools
- Lowest: Virgin Islands (VI) with 9 schools

5 Question 4: Full-Time Teachers Distribution

5.1 Question

Plot the distribution of Schools' full-time teachers over states and indicate which has the highest and lowest of High and Elementary schools.

5.2 Data Preprocessing

- Filtered out invalid FT_TEACHER values (negative sentinel codes, e.g., -999).
- Aggregated total FT_TEACHER by STATE and LEVEL_.
- Pivoted to wide format for HIGH and ELEMENTARY; filled missing with 0.
- Mapped state abbreviations to full names and merged with US_State.shp for geospatial plotting.
- Computed max/min states to annotate extremes on the maps.

5.3 Visualization

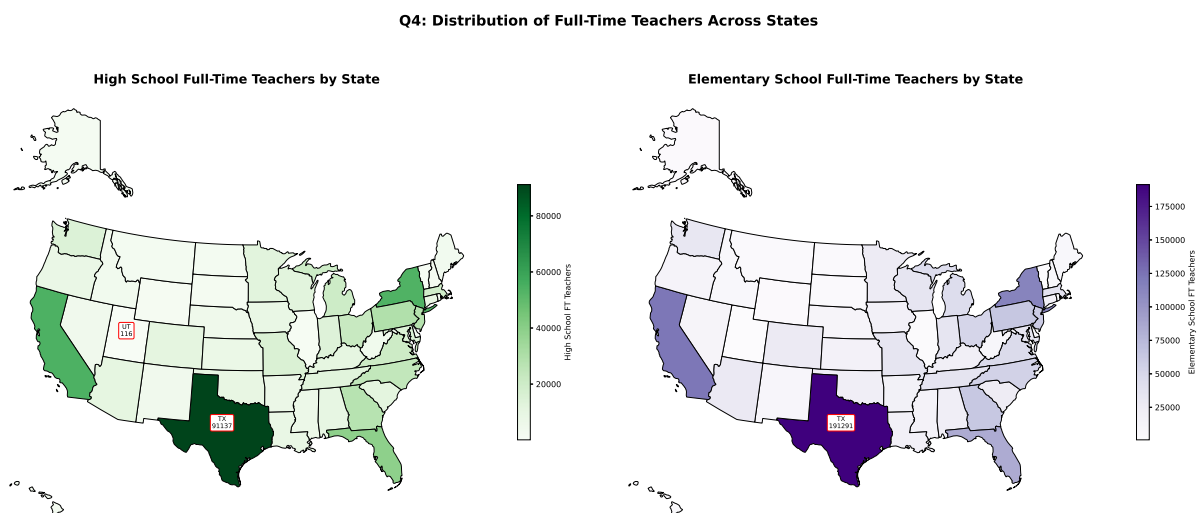


Figure 4: High School Full-Time Teachers (left) and Elementary School Full-Time Teachers (right) by State

5.4 Key Findings

- **High School Teachers:**
 - Highest: Texas (TX) with 91,137 teachers
 - Lowest: Utah (UT) with 116 teachers
- **Elementary School Teachers:**
 - Highest: Texas (TX) with 191,291 teachers
 - Lowest: Virgin Islands (VI) with 285 teachers

6 Question 5: Enrollment Distribution & Teacher-Student Ratio

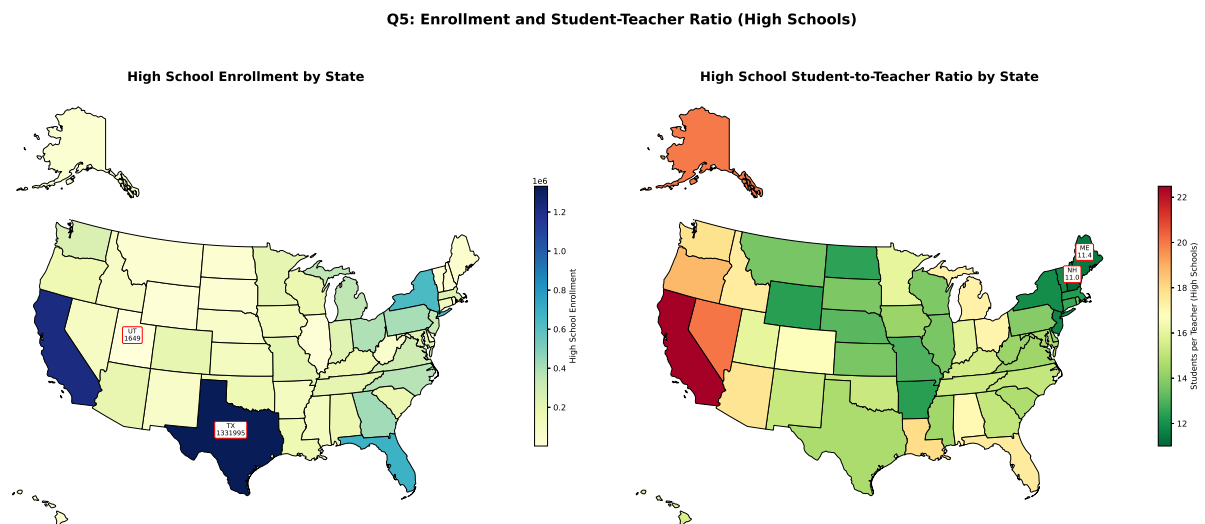
6.1 Question

Plot the distribution of Schools' Enrollment and compare it with the plots in Q3 and Q4. Indicate those three regions which have lowest Enrollment to Full-Time Teachers ratio.

6.2 Data Preprocessing

- Filtered to valid records: $\text{ENROLLMENT} \geq 0$ and $\text{FT_TEACHER} \geq 0$ (exclude sentinels like -999).
- Aggregated ENROLLMENT and FT_TEACHER totals by STATE and LEVEL_ .
- Computed student-to-teacher ratio as $\text{ENROLLMENT}/\text{FT_TEACHER}$ per state/level and selected the HIGH level for mapping.
- Mapped state abbreviations to names and merged with `US_State.shp` for choropleths.

6.3 Visualization



6.4 Key Findings

- **Highest Enrollment State (High Schools):** Texas (TX) with **1,331,995** students
- **Lowest Enrollment State (High Schools):** Utah (UT) with **1,649** students
- **Three States with Lowest Student-Teacher Ratios (High Schools):**
 - New Hampshire (NH): **11.00** students/teacher
 - Virgin Islands (VI): **11.08** students/teacher
 - Maine (ME): **11.37** students/teacher

7 Question 6: Global Corruption and Economic Development Analysis

7.1 Background

Corruption remains one of the most significant barriers to sustainable development, economic growth, and social justice worldwide. This analysis examines the Transparency International Corruption Perceptions Index (CPI) alongside GDP per capita data to explore the relationship between corruption and economic prosperity.

7.2 Question

Using the Kaggle corruption index dataset and World Bank economic indicators, plot significant visualization maps to describe the relationship between global corruption patterns and economic development.

Data Sources:

- Corruption Perceptions Index: <https://www.kaggle.com/datasets/transparencymint/corruption-index>
- GDP per Capita (2016): World Bank Development Indicators via public API

7.3 Data Preprocessing

The following preprocessing steps were applied to prepare the integrated dataset:

1. **Corruption Data:** Downloaded CPI 2016 scores from Kaggle dataset
2. **GDP Data:** Retrieved 2016 GDP per capita (current US\$) from World Bank API
3. **Data Filtering:** Removed aggregate regional codes, keeping only valid ISO3 country codes (3-letter alphabetic)
4. **Geographic Merging:** Matched both indicators to Natural Earth world boundaries via ISO_A3 codes
5. **Normalization:**
6. Scaled **GDP per capita** to 0-100 range using min-max normalization for visual comparability
7. **Color Semantics:** Unified color scheme across both maps (red = high/worse, green = low/better)

7.4 Visualization

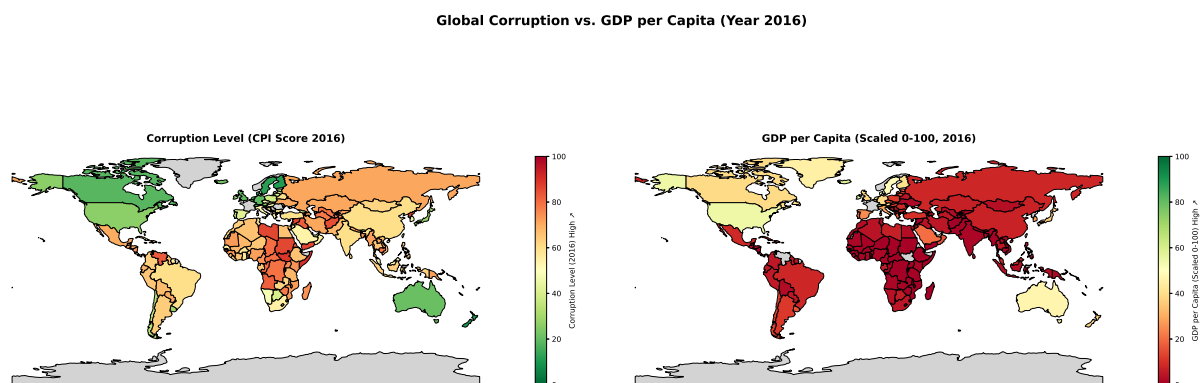


Figure 6: Global Corruption vs. GDP per Capita (Year 2016)

This side-by-side choropleth visualization presents two key indicators on comparable 0-100 scales:

- **Left Panel - Corruption Level:** Derived from CPI score, where higher values indicate more corruption (red) and lower values indicate cleaner institutions (green)
- **Right Panel - GDP per Capita (Scaled):** Min-max normalized to 0-100, where higher values (green) indicate greater economic prosperity and lower values (red) indicate poverty

Visual Patterns:

- **Green regions in both maps** (Nordic countries, North America, Australia): Strong correlation between low corruption and high GDP
- **Red regions in both maps** (Africa, parts of Asia): Systemic corruption paired with low economic development

7.5 Key Findings

- **Coverage:** 156 countries with corruption data, 163 countries with GDP data (2016)
- **GDP Range:** \$235 - \$106,899 per capita (normalized on 0-100 scale)
- **Corruption Extremes:**
 - Cleanest: New Zealand (CPI 90, Corruption Level 10)
 - Most Corrupt: Somalia (CPI 10, Corruption Level 90)
 - Global CPI Average: 42.9
- **Correlation Insight:** Visual inspection reveals a strong inverse relationship where countries with lower corruption (green, left) tend to have higher GDP per capita (green, right), suggesting that corruption significantly impedes economic development
- **Regional Patterns:**
 - **Developed Economies:** Nordic countries, North America, and Australia exhibit both low corruption and high GDP
 - **Developing Regions:** Sub-Saharan Africa and parts of Central Asia face dual challenges of high corruption and low economic output
 - **Emerging Markets:** Mixed results where some show improving corruption scores alongside growing GDP